Simulation study of honest tensor-functional intervals

Status: PASSED; completed simulation results, 2026-09-13. Real-data execution remains pending a dataset and justified sampling model. The open theoretical problems are unchanged.

The study executed **1,084,640 Monte Carlo data sets** in **318 stochastic configurations**, plus **84 analytic-only configurations** (402 statistical/envelope configurations total). A separate 18-setting runtime study is excluded from these coverage counts. The initial registry had 351 configurations and 824,640 data sets; a separately registered requirements/calibration follow-up added 51 configurations and 260,000 data sets. There are 2,160 binary optimizer solves on 240 shared data sets, not that many independent coverage observations. All summaries, raw per-trial endpoints/errors, seeds and numerical statuses are retained.

The main findings are rate-scale shortening in the regimes with executable guarantees, a substantial precision cost for conservative pointwise correction, severe failures under incorrectly supplied assumptions, and remaining computational gaps. Simulations do not prove minimax optimality, uniform honesty, or impossibility.

Design and checks

The design, theorem map and initial registry were written before implementation/execution. Gaussian sample-mean experiments use tau=1/sqrt(n). Scalar/diagonal sufficient observations are generated exactly for methods that use only those contractions; full tensors are generated for projection and optimization. Procedures share observations, and split ablations share the same valid artificial Gaussian split. Supplied bounds are not selected using truth after seeing a realization. Signal construction uses truth only to define the prescribed parameter class and explicitly labeled oracles.

The study uses 329 known-answer checks, including 36 equivalence checks against the preserved cap-valid implementation, exact Tucker tangent-Jacobian variance checks, noiseless reconstruction, net calibration and model membership. Final analysis independently recomputes all interval coverages and mean widths, verifies all configuration/source hashes and preserves 330 earlier files unchanged. Python/NumPy/SciPy versions and single-thread settings are in the run manifests.

Suite	Registered cases	MC data sets	Summary rows
S01	54	270,000	486
S02	44	44,000	660
S03	8	2,900	144
S04	28	14,000	588
S05	10	50,000	60
S06	15	7,500	90
S07	168	420,000	504
S08	8	240	120
S09	11	11,000	99
S10	5	5,000	90
S12	50	250,000	600
S13	1	10,000	3

Cheap cases use B=5,000, giving nominal 95% coverage MCSE .00308. Full-tensor B is 500, 300 or 150, with nominal MCSE .00975, .01258 or .01780. Solver B=30 gives .03979. Wilson intervals are retained even at 0/100% coverage. Mean lengths, losses and paired differences have uncertainty summaries. Squared-loss Monte Carlo standard errors are not claimed for t3 noise, whose fourth moment is infinite. Aggregate timing from the prolonged host session includes scheduling/inactive periods; separate S11 calls provide the computational comparison.

Coverage and rate validation

The table gives descriptive ranges over tested parameters, not confidence bounds for worst-case coverage over a continuum. All 90%, 95% and 99% levels are in the full tables. The predeclared family-wise lower-coverage screen includes 2526 in-scope rows and flagged 0 at Bonferroni cutoff 1.98e-05. Lack of a flag is not proof of uniform coverage and does not hide the individual low-coverage case discussed below.

Suite	Method	In-scope cases	95% coverage range	B per case
S01	growing	54	0.944–1.000	5000–5000
S02	threshold_selected	44	1.000–1.000	1000–1000
S03	pointwise	8	0.920–0.993	150–500
S04	pointwise	26	0.940–0.986	500–500
S05	calibrated	9	0.949–1.000	5000–5000
S06	directional_mom	12	1.000–1.000	500–500

Curve	Observed slope	MC slope SE	Reference
G1 fixed-M structural length vs p	-0.2500	deterministic	-0.25
G1 noise-branch length vs n	-0.5000	deterministic	-0.5
G1 noise-branch RMSE vs n	-0.4980	0.0023	-0.5
P1 length vs p: cap 1	-0.4745	0.0053	-0.5
P1 length vs p: cap ceil(sqrt(p))	-0.1791	0.0071	-0.25
Threshold raw length vs p, K=3 (log factor retained)	-0.1526	deterministic	-0.1667
Threshold raw length vs p, K=4 (log factor retained)	-0.2289	deterministic	-0.25

Log-regression slopes are descriptive finite-grid diagnostics. Mean-length error bars vanish for deterministic-radius procedures, so their fitted slopes partly check formulas, not stochastic attainment. The noise-branch RMSE and full-tensor expected-length slopes use observed losses/lengths with Monte Carlo uncertainty. Threshold reference exponents exclude the displayed logarithmic correction. See `tables/summary.csv` for length/rate, squared-loss/rate-squared and paired differences.

The P1 slopes do not equal their limiting references over p=8–64: the fixed-cap slope is about -0.474 and the growing-cap slope about -0.179. These departures exceed the reported Monte Carlo slope errors, which do not account for finite-dimension curvature. We do not label this exact exponent validation. A post-hoc diagnostic freezes the fitted subspaces at their true values and the singular values at the prescribed amplitude. If the resulting tangent and direct radii are h and d, their center difference is N(0,tau^2), and the intersection length is max(0,min(2min(h,d),h+d-abs(W))). Integrating this expression predicts the finite-dimension clipping effect. All eight approximate mean-length predictions are within two observed Monte Carlo standard errors; the integral formula is independently checked by quadrature. This is an explanatory approximation using unavailable true geometry, not a new theorem for estimated subspaces.

p	Cap	Observed length / rate	Frozen-subspace length / rate	Difference in MCSE
8	1	7.578	7.478	1.95
8	3	6.271	6.310	-0.93
16	1	7.807	7.770	0.75
16	4	6.840	6.877	-0.69
32	1	7.857	7.908	-0.74
32	6	7.038	7.175	-1.77
64	1	8.009	7.974	0.43
64	8	7.277	7.476	-1.50

Pointwise precision and an estimator failure

p	Cap	Method	95% coverage [Wilson CI]	Mean length ± MCSE
16	1	direct	0.962 [0.941, 0.976]	3.92 ± 0
16	1	pointwise	0.982 [0.966, 0.991]	1.95 ± 0.012
16	1	xch_known	0.944 [0.920, 0.961]	0.98 ± 0
16	1	oracle	0.944 [0.920, 0.961]	0.98 ± 0
16	4	direct	0.962 [0.941, 0.976]	3.92 ± 0
16	4	pointwise	0.978 [0.961, 0.988]	3.42 ± 0.027
16	4	xch_known	0.954 [0.932, 0.969]	1.96 ± 0
16	4	oracle	0.954 [0.932, 0.969]	1.96 ± 0
64	1	direct	0.960 [0.915, 0.982]	3.92 ± 0
64	1	pointwise	0.993 [0.963, 0.999]	1 ± 0.0098
64	1	xch_known	0.960 [0.915, 0.982]	0.49 ± 0
64	1	oracle	0.960 [0.915, 0.982]	0.49 ± 0
64	8	direct	0.873 [0.811, 0.917]	3.92 ± 0
64	8	pointwise	0.920 [0.865, 0.954]	2.57 ± 0.047
64	8	xch_known	0.927 [0.873, 0.959]	1.39 ± 0
64	8	oracle	0.927 [0.873, 0.959]	1.39 ± 0

At p=16, cap one, the P1 test amplitude is about 310,930 noise standard deviations. Such signals deliberately satisfy a conservative sufficient condition; practical signal sweeps extend from zero upward and show that this condition is far from a useful empirical cutoff. The Xu–Chen–Han comparator uses its two simultaneous iterations and exact fitted-core tangent variance. It is an asymptotic regular-signal procedure, not an honest weak-signal competitor, and often gives materially shorter intervals in the strong regime. The support oracle has unavailable structure. There is no claim of leading-constant or universal practical dominance.

The cap-valid implementation replaces empty interval intersections by {0}. At large nonzero means its **interval midpoint can have enormous MSE**: in S03_002, the observed midpoint MSE is 1.20847e+07 (MCSE 1.21e+07), despite coverage 0.982 and mean length 1.952. The large squared-loss uncertainty reflects rare sentinel events. There are 16 tested 95% pointwise cases with at least one such nonzero-target sentinel. This is retained in raw records and the audit. The P1 theorem concerns honesty and expected length; it does not promise midpoint MSE control. A useful estimation repair needs a separate analysis, rather than silently reinterpreting this midpoint as the ideal residual-fit estimator.

Assumption failures and calibration investigation

Under equicorrelated Gaussian coordinates with rho=.8, naive iid calibration covers 0.1094, while the known-covariance interval covers 0.9512. Mean-shift contamination gives naive coverage 0.0034; a variance correction does not remove its bias. The finite-variance median scan tests the full binary net and a known covariance upper bound. Its simultaneous feasibility is conservative in these draws; the study does not implement the general robust minimax projection and does not claim dependence or Huber-contamination robustness.

Understated rank caps produce minimum pointwise coverage 0.000 in the deliberate violations. The previous rank selector can return a zero interval at weak nonzero signals, outside its separation assumption. Loading residual and approximation-allowance omissions also cause severe failures. Corrected intervals retain their explicitly supplied assumptions; general arbitrary-loading and approximate-rank sharp computation is not established by these examples.

The original S03_007 direct Gaussian interval covers 0.873 [0.811, 0.917] with B=150. This unusually low realization is displayed, including its interval excluding .95; it is not discarded because the broad family-wise screen did not flag it. Regeneration of its full Gaussian tensors checks the direct statistic independently. The separately registered S13 confirmation covers 0.949 [0.945, 0.953] with B=10,000, using a new fixed stream of exact diagonal observations. It is a diagnostic addition, not a replacement or a pooled result. See `results/gaussian_diagnostic.json` for the original one-sided test and regeneration discrepancy.

Computation and intermediate caps

Signal	Loading	Certified at budget 3000	Mean finite gap	No witness
zero	flat	30/30	0.05858	0
zero	arbitrary	30/30	0.04044	0
weak	flat	30/30	0.09386	0
weak	arbitrary	0/30	0.1446	0
boundary	flat	26/30	0.1009	0
boundary	arbitrary	30/30	0.09998	0
strong	flat	0/30	0.1444	0
strong	arbitrary	0/30	0.2093	0

Only continuous rank-one binary tensors are optimized. All support levels in the implemented projective net are included. For every tolerance/budget the outer interval is retained, along with feasible witnesses and endpoint gaps; unmet tolerances are explicit. Float64 guard bounds are numerical evidence, not formal interval arithmetic. The ideal general-dimensional minimax residual/block optimizer remains absent. S11 reports actual diagonal access through p=1,048,576, full tensors through 64^3, and binary order-three/four costs; traced allocations exclude untraced native workspace.

Cap at p=65536	Executable length	Diagonal-mixture lower	Upper / lower
1	2	0.007709	259
16	3.92	0.0809	48.5
64	3.92	0.2009	19.5
256	3.92	0.2993	13.1
1024	3.92	0.3155	12.4

The follow-up uses sparse common-target diagonal signals. Its diagonal-mixture lower bound need not be the strongest available fixed-cap bound, especially for K=3. The remaining upper/lower gap is a diagnostic of these constructions, not a new minimax theorem. Matching intermediate growing-cap factors, arbitrary loading profiles and a complete pointwise characterization remain open.

Local power and distribution diagnostics

Intervals are inverted for the zero functional at strong signed rank-two means. Alternatives shift the target by c tau sqrt(r/p), with c in {0,1,2,4,8}. The exact direct-Gaussian power curve and empirical paired rejection probabilities are reported. Direct and oracle statistics are exact Gaussian pivots; XCH diagnostics are interpreted under regular-signal assumptions. No CLT is asserted for nonlinear interval midpoints or global-solver endpoints.

Local shift	Method	Rejection probability [Wilson CI]
0.0	direct	0.059 [0.046, 0.075]
0.0	pointwise	0.031 [0.022, 0.044]
0.0	xch_known	0.048 [0.036, 0.063]
0.0	oracle	0.048 [0.036, 0.063]
1.0	direct	0.070 [0.056, 0.088]
1.0	pointwise	0.049 [0.037, 0.064]
1.0	xch_known	0.151 [0.130, 0.175]
1.0	oracle	0.151 [0.130, 0.175]
2.0	direct	0.108 [0.090, 0.129]
2.0	pointwise	0.130 [0.111, 0.152]
2.0	xch_known	0.496 [0.465, 0.527]
2.0	oracle	0.496 [0.465, 0.527]
4.0	direct	0.278 [0.251, 0.307]
4.0	pointwise	0.565 [0.534, 0.595]
4.0	xch_known	0.984 [0.974, 0.990]
4.0	oracle	0.984 [0.974, 0.990]

Local shift	Method	Rejection probability [Wilson CI]
8.0	direct	0.827 [0.802, 0.849]
8.0	pointwise	0.998 [0.993, 0.999]
8.0	xch_known	1.000 [0.996, 1.000]
8.0	oracle	1.000 [0.996, 1.000]

Real-data status and outstanding empirical requirements

No scientifically justified repeated-tensor dataset, independent target loading or known directional noise calibration was supplied. `real_data/README.md` records the missing sampling units, calibration, rank/approximation bounds, missingness and split/preprocessing requirements. R01 remains pending. Simulated observations, synthetic tensors and noise-injected images are not represented as real-data validation.

The main empirical limitations are the small solver replication count, conservative P1 signal floor, largest-p coverage uncertainty, unavailable general residual/block fits, unmatched intermediate-cap bounds, and lack of a justified application. Lower bounds and impossibility results are analytic statements, not experimentally proved. The full reviewer audit and theory/experiment comparison separate those gaps from completed simulation execution.

Figures

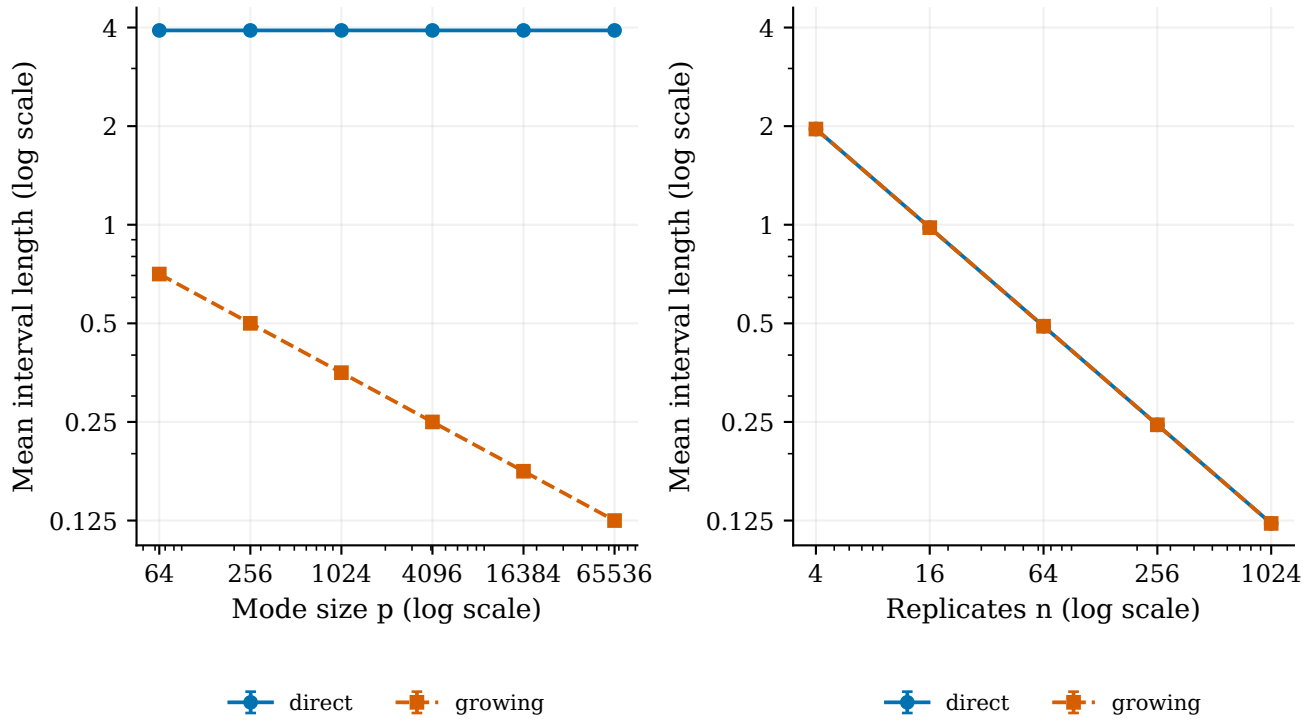


Figure 1: The executable large-cap interval follows its structural and noise branches. Left: M=1, tau=1, r=sqrt(p), boundary diagonal signals. Right: p=1024, r=32, M=8, tau=n^(-1/2), n>=4. Each point uses 5,000 data sets; interval lengths are deterministic, so mean-length error bars vanish. The tested exponents are -1/4 in p and -1/2 in n.

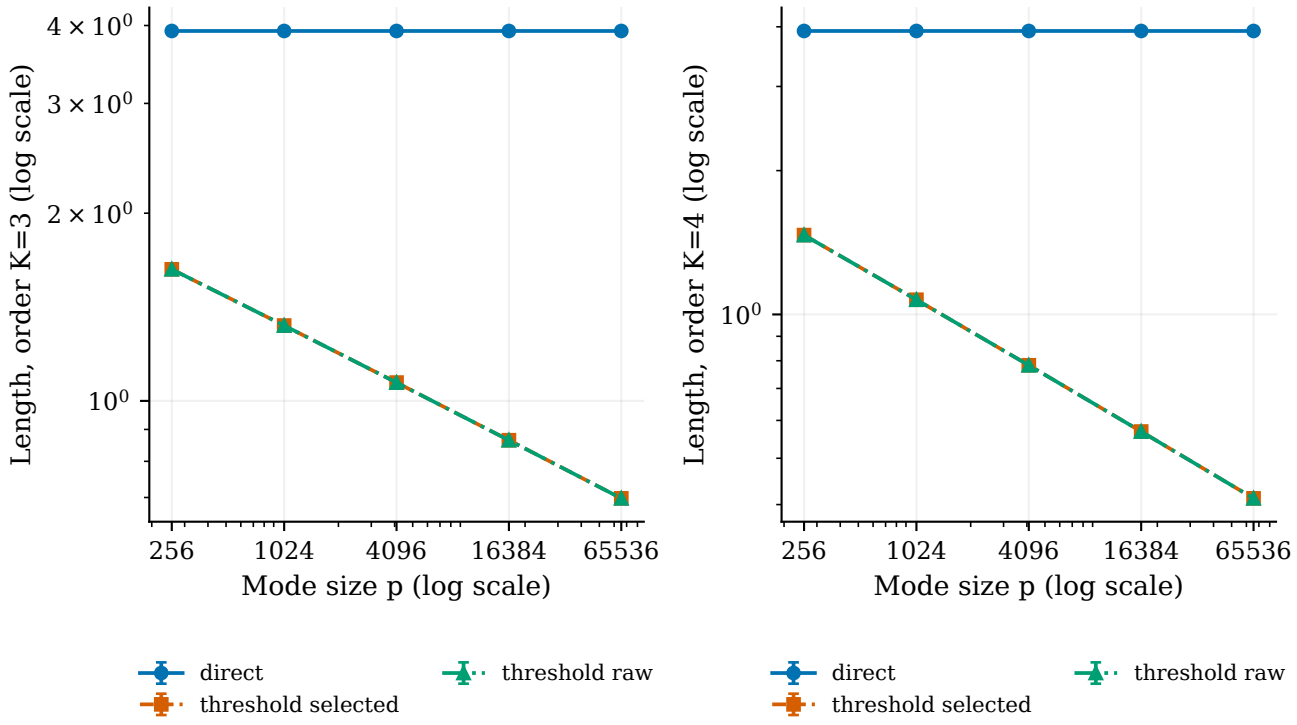


Figure 2: The fast diagonal interval becomes shorter as dimension grows, with conservative coverage in the tested signals. M=sqrt(p), tau=1, rank-one support near the threshold boundary; 1,000 data sets per configuration. The raw threshold curve differs from deterministic selection when structural or direct intervals are shorter. This evaluates the diagonal procedure, not attainment of the sharper full-data ideal rate.

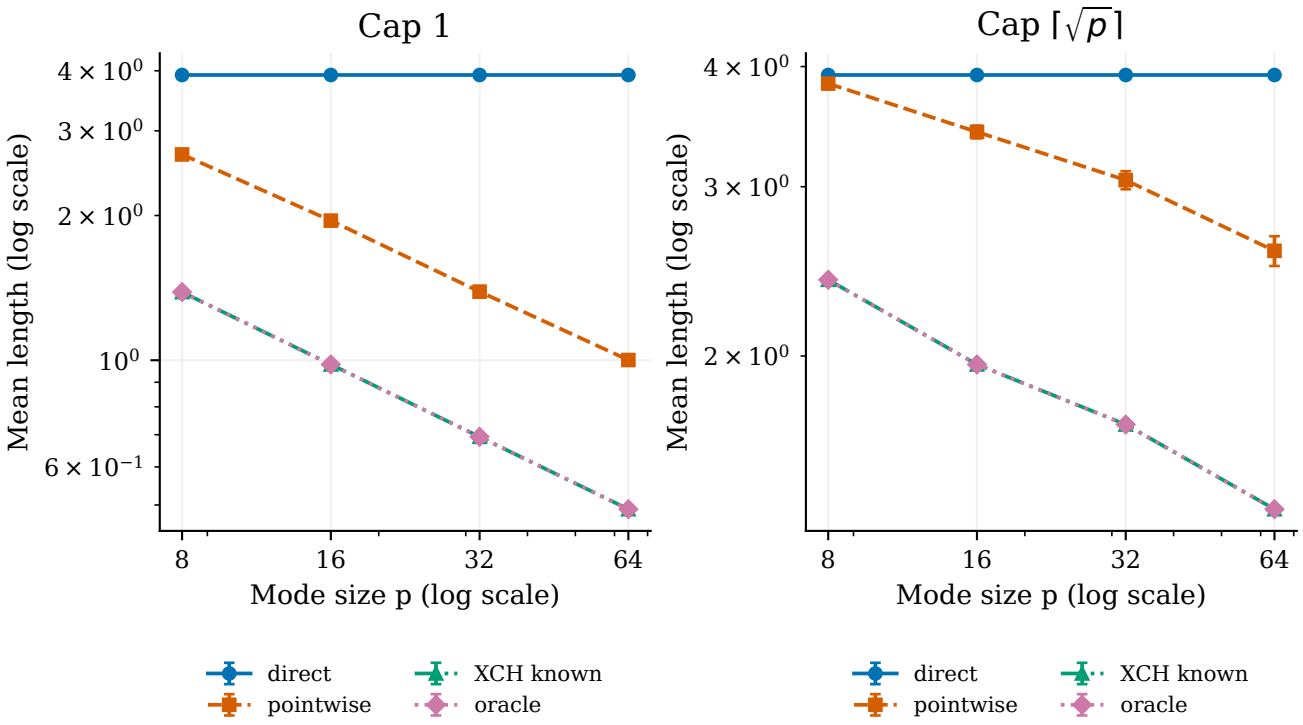


Figure 3: Strong saturated signals allow substantial shortening, while the regular-signal comparator and support oracle can be shorter than the finite-sample certificate; their curves nearly coincide here. Signal amplitudes equal eight times the P1 sufficient bound. Replications are 500 at p=8/16, 300 at p=32, and 150 at p=64; bars are 1.96 mean-length standard errors. XCH denotes the locally reimplemented Xu–Chen–Han PCA procedure with known noise. Pointwise optimality is a rate claim, not leading-constant dominance.

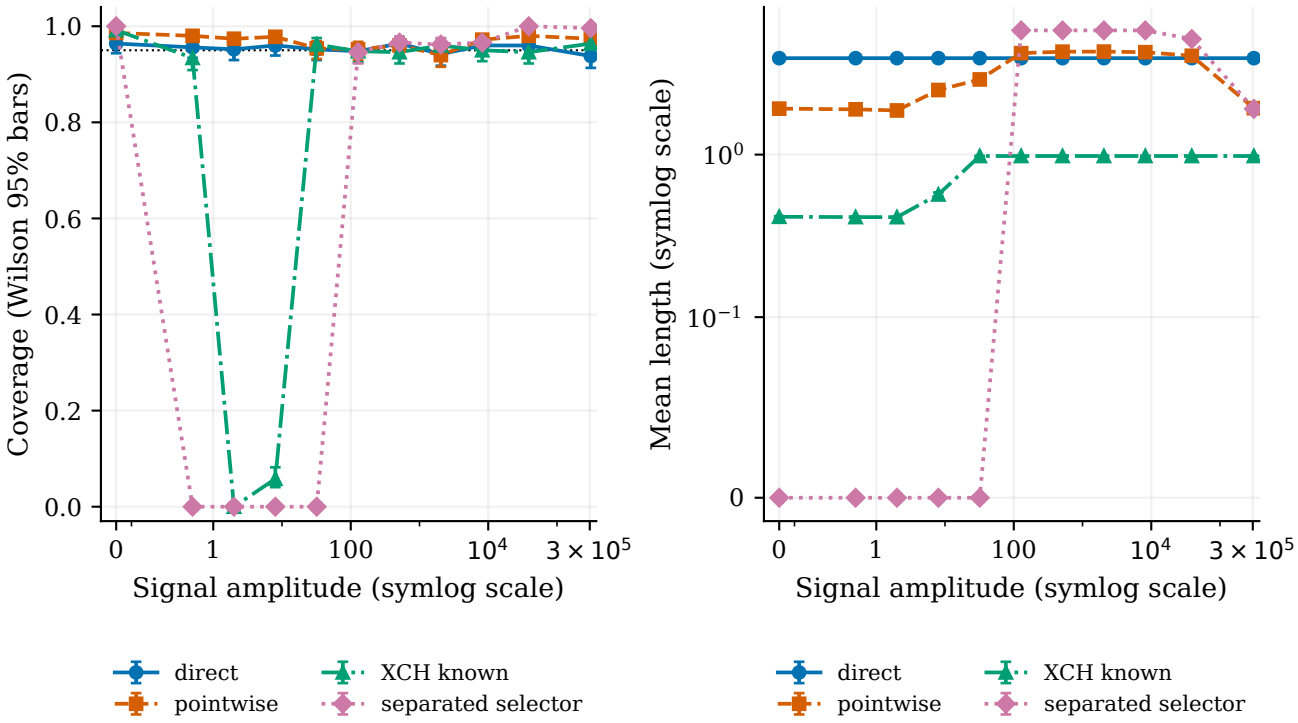


Figure 4: The separated rank selector fails for weak nonzero components, and the cap-valid correction can remain wider than direct inference well into the signal sweep. p=16, supplied cap one, 500 paired data sets per amplitude; confidence level 95%. Coverage bars are Wilson intervals and length bars are 1.96 mean standard errors. Weak-signal results are outside the regular/separated comparators' guarantees. Symlog axes retain zero amplitudes and zero reported widths.

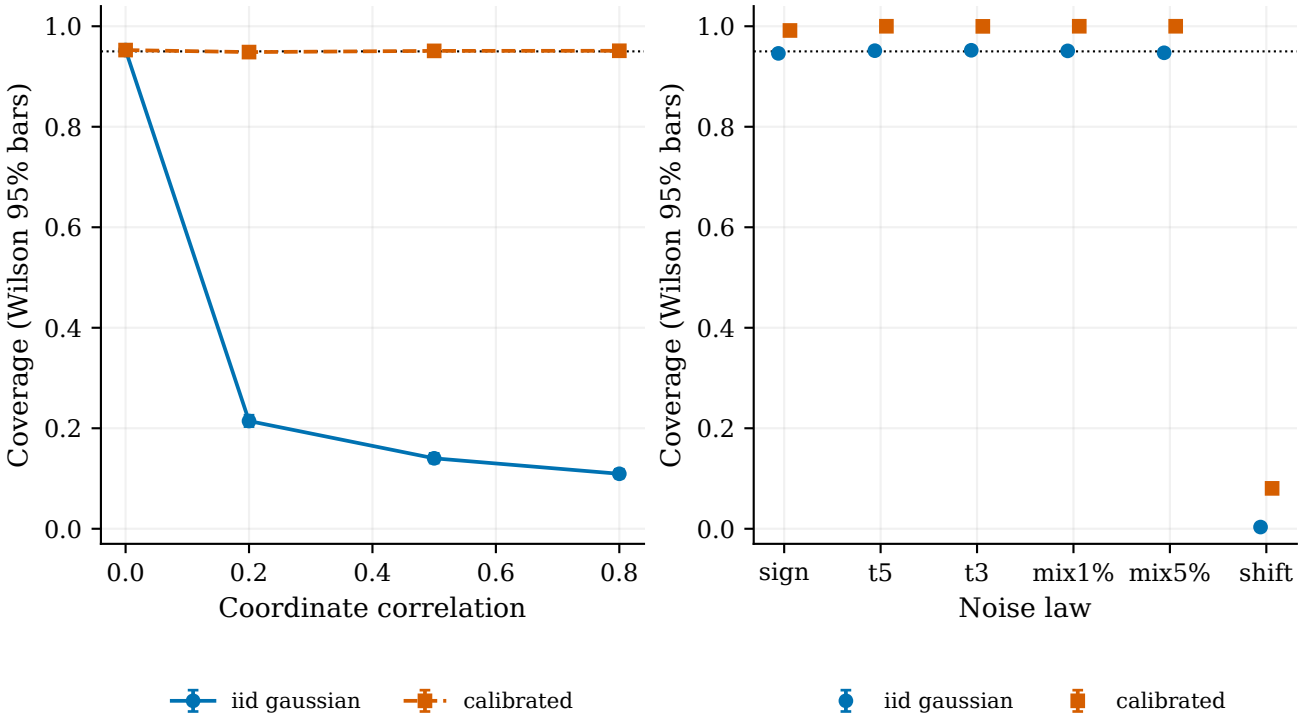


Figure 5: Correct covariance calibration restores coverage under correlated Gaussian coordinates; a mean shift is not repaired by variance correction. p=256 and 5,000 data sets per noise law. The calibrated method uses exact Gaussian covariance, a known sub-Gaussian proxy for bounded/mixture laws, or a Chebyshev variance bound for t3/t5. Both mean-shift curves deliberately ignore the nonzero noise mean and have no centered-noise guarantee. Mixtures are variance-standardized.

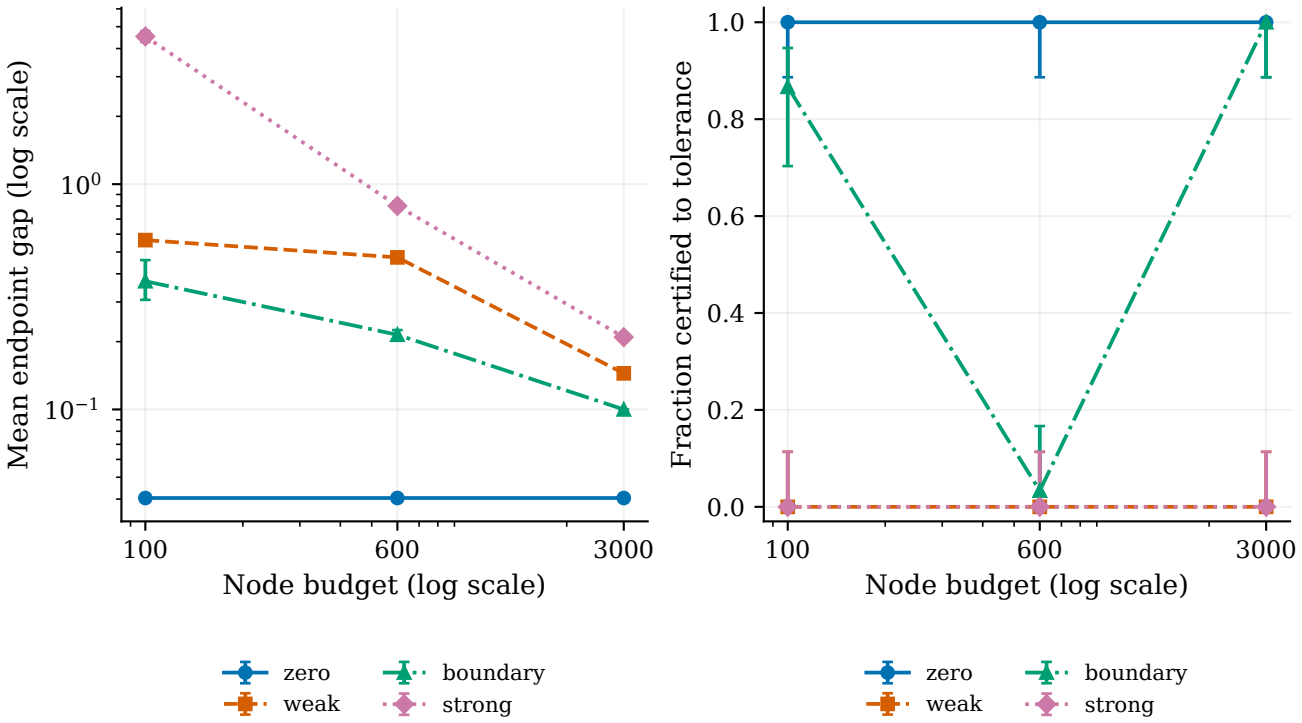


Figure 6: Finite search budgets can leave large endpoint gaps for arbitrary binary loadings. K=3, mode size two, 30 shared observations per signal/loading case; gap bars use 2,000 fixed-seed bootstrap resamples and certification bars are Wilson 95% intervals. Gap averages condition on a finite feasible-witness gap; missing-witness counts are in the table. Each signal case has its own independently fixed arbitrary loading. The budget-100 tolerance is .5 tau; budgets 600 and 3000 use .1 tau, so the first certification point uses a looser tolerance. Gaps are numerical float64 bounds with guards, not formal interval arithmetic.

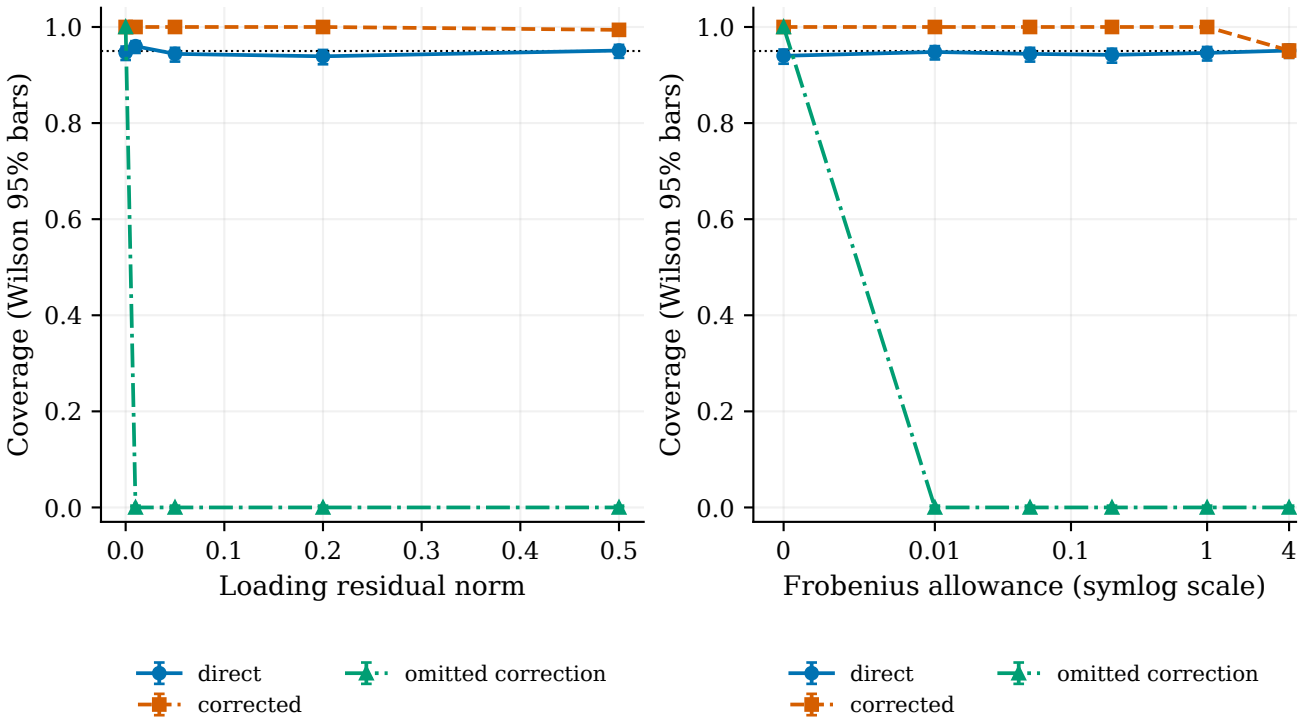


Figure 7: Omitting known loading or approximation contributions can destroy coverage. p=4096, K=4, 1,000 paired data sets per point. Left: rank-one off-diagonal mean, M=64, loading a flat diagonal plus a known off-diagonal product atom. Right: a rank-one core of norm one plus eta times the unit flat loading. These are executable conservative corrections, not the unavailable sharp full-data optimizer. Bars are Wilson 95% intervals.

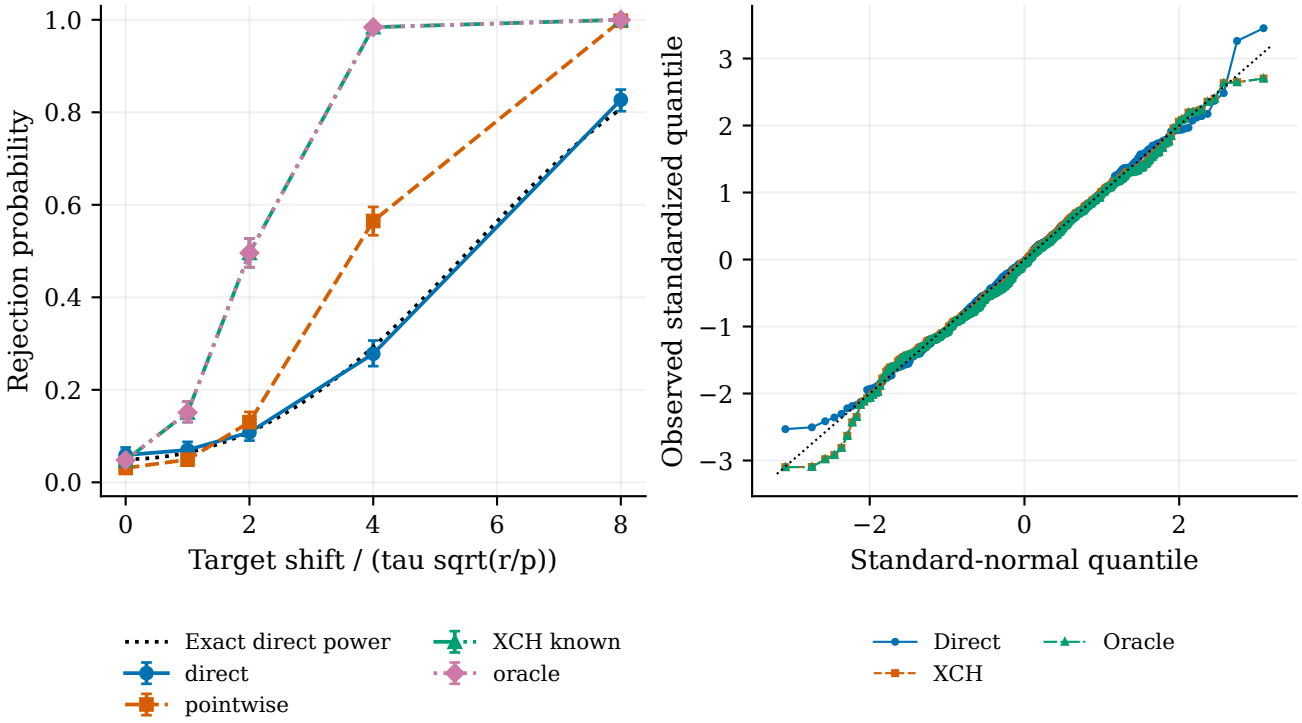


Figure 8: Regular-signal intervals and the support oracle can resolve smaller target shifts than the direct interval. Left: p=16, cap two, opposite large diagonal amplitudes, 1,000 draws per local alternative; bars are Wilson 95% intervals and the dotted curve is exact direct-Gaussian power. Right: the strong p=16 cap-one case, 500 draws, with exact direct/oracle pivots and the XCH regular-signal diagnostic. No normality claim is made for the cap-valid interval midpoint.

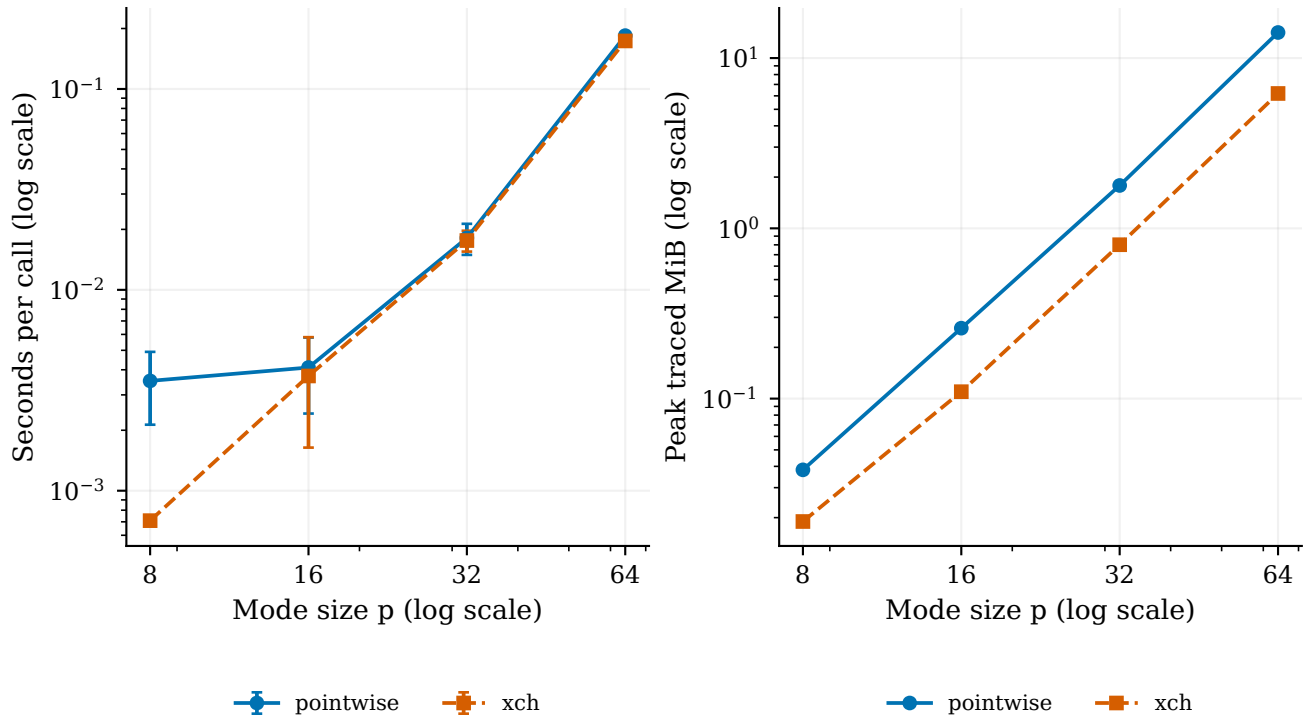


Figure 9: Full-tensor projection costs grow with the explicit input size. Fixed thread count, one warmup and ten measured calls per point; runtime bars are descriptive plus/minus one standard error, not 95% confidence intervals. The wider symmetric 1.96-SE interval crosses zero in one case and cannot be displayed fully on a log axis; all mean/SE values remain in the table. Memory counts traced Python/NumPy allocations during a call, excluding preallocated input and untraced native workspace; it is not total process memory. XCH and cap-valid calls both reuse factors across confidence levels. Separate tables report actual O(p) diagonal access and binary K=3/4 optimizer costs.

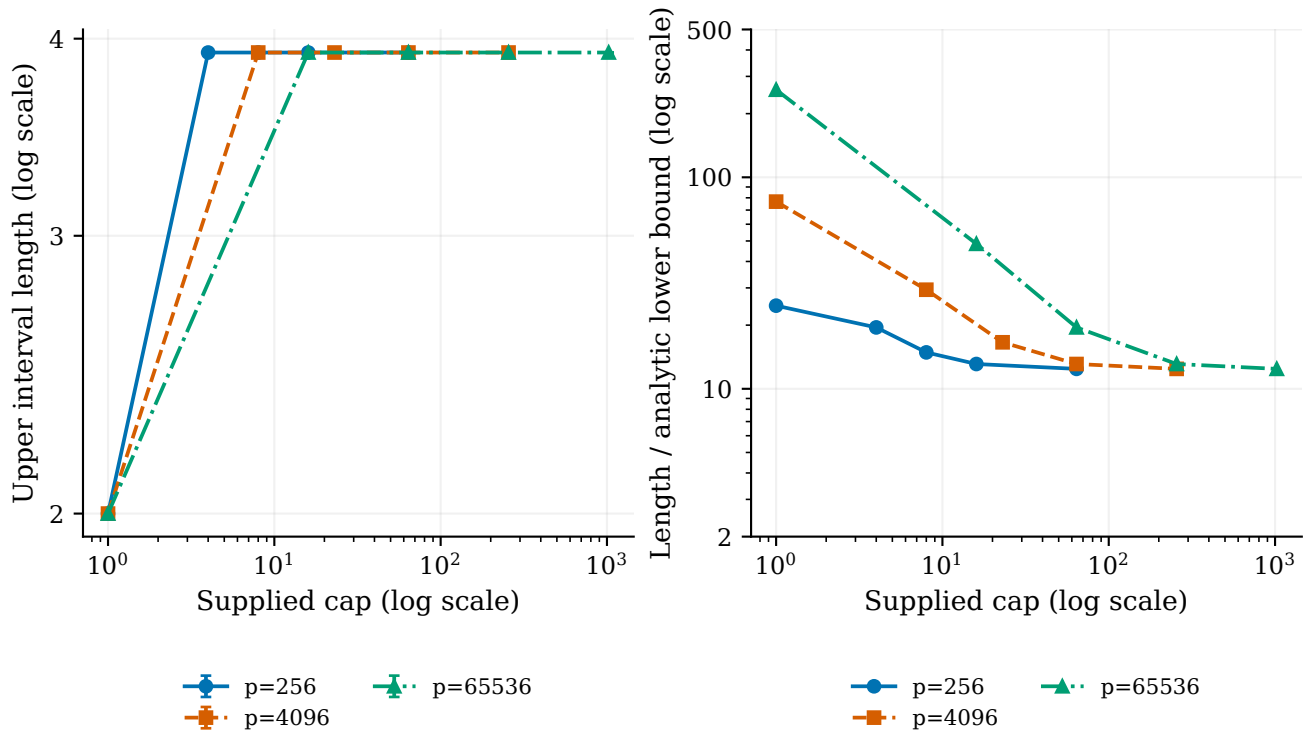


Figure 10: The valid cap-based upper interval leaves a substantial upper/lower gap at small and intermediate caps. This follow-up was registered separately after a requirements audit: 5,000 draws per configuration, sparse common-target diagonal means, M=sqrt(p), tau=1. The lower curve is an analytic testing bound, not an empirically attained benchmark. Matching intermediate-cap rates remain open. Interval lengths here are deterministic.